MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – CHAPTER TEST (SET – 4)** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 7: Coordinate Geometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units where applicable.
2. In alternative (**OR**) questions, complete solutions and step marking are provided for both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. The perimeter of the triangle formed by the vertices $(0,0)$, $(2,0)$ and $(0,2)$ is: [1]

(A) 4 units **(B)** 6 units **(C)** $6\sqrt{2}$ units **(D)** $(4+2\sqrt{2})$ units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) $(4+2\sqrt{2})$ units**

Given: Vertices $O(0,0)$, $A(2,0)$, and $B(0,2)$.

Formula Used: Perimeter $= OA + OB + AB$, $\quad d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$$OA = 2 \text{ units}, \quad OB = 2 \text{ units}, \quad AB = \sqrt{(0-2)^2+(2-0)^2} = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2} \text{ units}$$

$$\text{Perimeter} = 2+2+2\sqrt{2} = \boxed{(\mathbf{4+2\sqrt{2}}) \text{ units}}$$

Q2. If the endpoints of a diameter of a circle are $(-4,2)$ and $(4,-6)$, then the diameter of the circle is: [1]

(A) $8\sqrt{2}$ units **(B)** 10 units **(C)** $4\sqrt{5}$ units **(D)** $10\sqrt{2}$ units

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) $8\sqrt{2}$ units**

Given: Endpoints of diameter $A(-4,2)$ and $B(4,-6)$.

Formula Used: Diameter $AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$$AB = \sqrt{(4-(-4))^2+(-6-2)^2} = \sqrt{8^2+(-8)^2} = \sqrt{64+64} = \sqrt{128} = \boxed{\mathbf{8\sqrt{2}} \text{ units}}$$

Q3. The ratio in which the x-axis divides the line segment joining the points $A(1,-3)$ and $B(4,5)$ is: [1]

(A) $3:5$ **(B)** $5:3$ **(C)** $1:4$ **(D)** $2:3$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) $3:5$

Given: $A(1,-3)$ and $B(4,5)$. Let the x-axis intersect AB at $P(x,0)$ in ratio $k:1$.

Formula Used: $y = \dfrac{m_1y_2 + m_2y_1}{m_1 + m_2}$

Equating the y-coordinate to 0:

$$\frac{k(5)+1(-3)}{k+1} = 0 \implies 5k-3=0 \implies k = \frac{3}{5} \implies \boxed{\text{Ratio } = \mathbf{3:5}}$$

Q4. Two of the vertices of $\triangle PQR$ are $P(-1,5)$ and $Q(5,2)$. The coordinates of a point which divides PQ in the ratio $2:1$ are: [1]

(A) $(3,-3)$ **(B)** $(5,5)$ **(C)** $(3,3)$ **(D)** $(5,1)$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) $(3,3)$

Given: $P(-1,5)$, $Q(5,2)$, ratio $m_1 : m_2 = 2:1$.

Formula Used: $(x,y) = \left(\dfrac{m_1x_2 + m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

$$x = \frac{2(5)+1(-1)}{2+1} = \frac{10-1}{3} = 3, \qquad y = \frac{2(2)+1(5)}{2+1} = \frac{4+5}{3} = 3 \implies \boxed{(\mathbf{3},\mathbf{3})}$$

Q5. If the point $C(k,4)$ divides the join of the points $A(2,6)$ and $B(5,1)$ in the ratio $2:3$, then the value of k is: [1]

(A) 16 **(B)** $\dfrac{28}{5}$ **(C)** $\dfrac{16}{5}$ **(D)** $\dfrac{8}{5}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (C) $\dfrac{16}{5}$

Given: $A(2,6)$, $B(5,1)$, dividing point $C(k,4)$, ratio $2:3$.

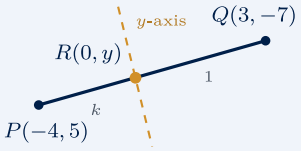
Formula Used: $x = \dfrac{m_1x_2 + m_2x_1}{m_1+m_2}$

Applying the formula for the x-coordinate k:

$$k = \frac{2(5)+3(2)}{2+3} = \frac{10+6}{5} = \boxed{\mathbf{\frac{16}{5}}}$$

Q6. In what ratio does the y-axis divide the line segment joining the points $P(-4,5)$ and $Q(3,-7)$? Also find the coordinates of the point of intersection. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Finding Ratio from x-Coordinate [1.5 Marks]
Given: $P(-4,5)$ and $Q(3,-7)$. Any point of intersection on the y-axis has abscissa 0, so take the point as $R(0,y)$ dividing PQ in ratio $k:1$.

We know the internal section formula:

Formula Used: $R(x,y) = \left(\dfrac{m_1x_2+m_2x_1}{m_1+m_2}, \dfrac{m_1y_2+m_2y_1}{m_1+m_2}\right)$

Equating abscissa to 0:

$$\frac{k(3)+1(-4)}{k+1} = 0 \implies 3k-4=0 \implies k=\frac{4}{3} \implies \text{Ratio} = \mathbf{4:3}$$

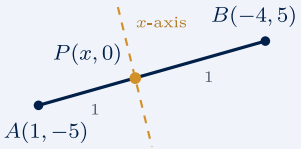
Step 2: Finding Coordinates of R [1.5 Marks]
Substituting $m_1=4$, $m_2=3$, $y_1=5$, and $y_2=-7$:

$$y = \frac{4(-7)+3(5)}{4+3} = \frac{-28+15}{7} = -\frac{13}{7}$$

$$\boxed{\text{Ratio } = \mathbf{4:3} \quad \text{and} \quad \text{Point of intersection } = \left(\mathbf{0}, -\frac{\mathbf{13}}{\mathbf{7}}\right)}$$

Q7. Find the ratio in which the line segment joining $A(1,-5)$ and $B(-4,5)$ is divided by the x-axis. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]



Step 1: Assuming Ratio $k:1$ on x-Axis [1 Mark]
Given: $A(1,-5)$ and $B(-4,5)$. Every point on the x-axis has ordinate 0, so let $P(x,0)$ divide segment AB internally in the ratio $k:1$.

We know the section formula for the y-coordinate:

Formula Used: $y = \dfrac{m_1y_2+m_2y_1}{m_1+m_2}$

Step 2: Solving for Ratio [2 Marks]
Equating the ordinate to 0:

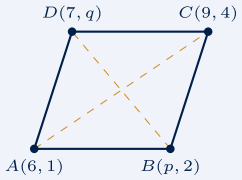
$$\frac{k(5)+1(-5)}{k+1} = 0 \implies 5k-5=0 \implies 5k=5 \implies k=1$$

(Also the point of division is $P = \left(\frac{1(-4)+1(1)}{2}, 0\right) = \left(-\frac{3}{2}, 0\right)$.)

$$\boxed{\text{Required ratio } = \mathbf{1:1}}$$

Q8. If the points $A(6,1)$, $B(p,2)$, $C(9,4)$ and $D(7,q)$ are the vertices of a parallelogram $ABCD$, then find the values of p and q. Hence, check whether $ABCD$ is a rectangle or not. [5]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Step 1: Finding p and q Using Diagonals [2.5 Marks]
Given: Parallelogram $ABCD$ with vertices $A(6,1)$, $B(p,2)$, $C(9,4)$, and $D(7,q)$. Since diagonals AC and BD of a parallelogram bisect each other, the midpoint of AC coincides with the midpoint of BD.

We know the midpoint condition for bisecting diagonals AC and BD:

Formula Used: $\left(\frac{x_A+x_C}{2}, \frac{y_A+y_C}{2}\right) = \left(\frac{x_B+x_D}{2}, \frac{y_B+y_D}{2}\right)$

Substituting the coordinates:

$$\left(\frac{6+9}{2}, \frac{1+4}{2}\right) = \left(\frac{p+7}{2}, \frac{2+q}{2}\right) \implies \left(\frac{15}{2}, \frac{5}{2}\right) = \left(\frac{p+7}{2}, \frac{2+q}{2}\right)$$

Equating abscissae and ordinates:

$$p+7=15 \implies \mathbf{p=8}, \qquad 2+q=5 \implies \mathbf{q=3}$$

Thus, $B(8,2)$ and $D(7,3)$.

Step 2: Checking Whether $ABCD$ is a Rectangle [2.5 Marks]
We know that a parallelogram is a rectangle **if and only if** its diagonals are equal ($AC = BD$). Using $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$:

$$AC = \sqrt{(9-6)^2+(4-1)^2} = \sqrt{3^2+3^2} = \sqrt{18} = 3\sqrt{2} \text{ units}$$

$$BD = \sqrt{(7-8)^2+(3-2)^2} = \sqrt{(-1)^2+1^2} = \sqrt{2} \text{ units}$$

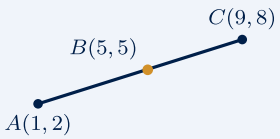
Since $AC \neq BD$ ($3\sqrt{2} \neq \sqrt{2}$), the diagonals are unequal.

$$\boxed{\mathbf{p=8}, \quad \mathbf{q=3}; \qquad ABCD \text{ is } \mathbf{not\ a\ rectangle}.}$$

Q9. Competency Case Study: On an annual athletics ground, markers are established at coordinates $A(1,2)$, $B(5,5)$, $C(9,8)$ and $D(5,5)$. [4]

(i) Find the distance between marker A and marker B.
(ii) Show that markers A, B and C lie in a single straight running lane.
(iii) Find the coordinates of the refreshment stall placed exactly midway along AC.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]



Given: Marker coordinates $A(1,2)$, $B(5,5)$, and $C(9,8)$ on the ground layout. We compute the distances AB, BC, and AC to check collinearity, and then apply the midpoint formula along AC.

We know the distance formula:

Formula Used: $d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

(i) Distance Between Marker A and Marker B: [1 Mark]

$$AB = \sqrt{(5-1)^2+(5-2)^2} = \sqrt{4^2+3^2} = \sqrt{16+9} = \sqrt{25} = \boxed{\mathbf{5}\text{ units}}$$

(ii) Proving A, B, C are Collinear (Straight Running Lane) [2 Marks]

Computing distances BC and AC:

$$BC = \sqrt{(9-5)^2+(8-5)^2} = \sqrt{4^2+3^2} = \sqrt{16+9} = \sqrt{25} = 5\text{ units}$$

$$AC = \sqrt{(9-1)^2+(8-2)^2} = \sqrt{8^2+6^2} = \sqrt{64+36} = \sqrt{100} = 10\text{ units}$$

Checking the collinearity condition:

$$AB + BC = 5 + 5 = 10 = AC$$

Hence, markers A, B, and C are **collinear** (lie in a single straight lane).

(iii) Refreshment Stall Midway Along AC [1 Mark]

Using the midpoint formula $M = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$ for $A(1,2)$ and $C(9,8)$:

$$M = \left(\frac{1+9}{2}, \frac{2+8}{2}\right) = \left(\frac{10}{2}, \frac{10}{2}\right) = \boxed{(\mathbf{5},\mathbf{5})}$$

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